Notes: Jaimovich and Rebelo (AER, 2009)

Can News about the Future Drive the Business Cycle?

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1 Big picture

- **Question.** Can news about future fundamentals generate a business-cycle boom today, even when current fundamentals have not yet changed?
- **Why the question is hard.** In a standard neoclassical model, good news about future productivity increases expected wealth. Households want more consumption and more leisure, so labor supply falls. Output can fall immediately: good news about tomorrow can create a recession today.
- **Paper's answer.** Yes, news can drive business-cycle comovement if the model contains three ingredients: variable capital utilization, adjustment costs to investment, and preferences with weak short-run wealth effects on labor supply.
- **Comovement target.** The model is designed to generate both *aggregate comovement*–output, consumption, investment, hours, and wages moving together–and *sectoral comovement*–output, labor, and investment rising together in consumption and investment sectors.
- **Shocks studied.** The fundamentals are aggregate total factor productivity (TFP), sectoral TFP in consumption and investment sectors, and investment-specific technological change.
- **Main result.** With the authors' preference specification near the GHH case, news about future productivity can trigger an expansion today. The model can also produce realistic-looking recessions without contemporaneous negative technology shocks: recessions are driven by disappointing information about the future.

2 Incremental contribution of the paper

- **Unified news-shock framework.** Earlier papers had shown isolated ways for expectations to matter. This paper constructs one model that can generate both aggregate and sectoral comovement in response to contemporaneous and anticipated shocks.
- **Preference bridge between KPR and GHH.** The paper introduces preferences that nest King-Plosser-Rebelo (KPR) preferences and Greenwood-Hercowitz-Huffman (GHH) preferences. The parameter γ governs the short-run strength of wealth effects on labor supply.
- **Mechanism rather than only calibration.** The paper explains precisely why the standard model fails and why the three additions work together: investment adjustment costs make agents act before the future shock arrives; variable utilization raises the marginal product of labor; weak short-run wealth effects prevent labor from falling.
- **Sectoral comovement.** A key contribution is the two-sector model. The paper shows that the same ingredients can overcome the Christiano-Fitzgerald sectoral-comovement problem when wealth effects are sufficiently weak.
- **News in business-cycle simulation.** The paper does not stop with impulse responses. It calibrates a stochastic model with investment-specific technical progress and noisy signals based on the Livingston survey, then compares model moments to US business-cycle moments.
- **Recession mechanism.** The paper provides a distinctive interpretation of recessions: downturns can arise not from negative productivity shocks today, but from the economy learning that

future investment-specific technical progress will be weaker.

3 Motivation and central puzzle

The paper begins from the observation that the late-1990s US investment boom looked tightly connected to optimistic expectations. The authors show that investment by S&P 500 firms is positively correlated with analysts' long-term earnings-growth forecasts but negatively correlated with realized earnings. This motivates a news-driven story: optimism about future technologies induces investment today; if those technologies fail to meet expectations, investment collapses.

Read: The empirical picture is suggestive rather than a structural test. The paper uses it to motivate the theoretical question: can a standard macro model turn optimism about future fundamentals into an expansion today?

3.1 Why standard models fail

The standard neoclassical model faces the Barro-King problem. With time-separable preferences, news of higher future productivity increases lifetime wealth. The household responds by consuming more leisure immediately, so hours worked decline. If current technology is unchanged, the fall in labor lowers output. Thus, the model tends to generate the wrong sign for hours and output in response to good news.

Economic message: The problem is not merely that news shocks are hard to measure. The deeper problem is that standard preferences and production choices imply the wrong substitution/wealth-effect balance.

4 Core model ingredients

4.1 Preferences and short-run wealth effects

The representative household maximizes expected discounted utility:

$$U = E_0 \sum_{t=0}^{\infty} \beta^t \frac{\left(C_t - \psi N_t^{\theta} X_t\right)^{1-\sigma} - 1}{1-\sigma},$$

where the habit-like auxiliary state satisfies

$$X_t = C_t^{\gamma} X_{t-1}^{1-\gamma}.$$

The parameter γ is central.

- When $\gamma = 1$, preferences reduce to the KPR-style specification with strong wealth effects.
- When $\gamma = 0$, preferences become GHH-style preferences with no wealth effect on labor supply.
- For $0 < \gamma < 1$, short-run labor-supply wealth effects can be weak while long-run preferences remain stationary.

Takeaway: The model needs weak *short-run* wealth effects, not necessarily zero wealth effects forever. This distinction is what the new preference specification buys.

4.2 Production and investment-specific technology

Output is produced with capital services and labor:

$$Y_t = A_t(u_t K_t)^{1-\alpha} N_t^{\alpha},$$

where A_t is aggregate TFP and u_t is capital utilization. Output can be used for consumption or for investment:

$$Y_t = C_t + \frac{I_t}{z_t}.$$

The variable z_t captures investment-specific technical change: a higher z_t means that a unit of output can be transformed into more investment goods.

4.3 Capital accumulation and investment adjustment costs

Capital evolves according to

$$K_{t+1} = I_t \left[1 - \varphi\left(\frac{I_t}{I_{t-1}}\right)\right] + [1 - \delta(u_t)]K_t.$$

Adjustment costs are incurred when investment growth changes. Utilization raises depreciation through $\delta(u_t)$, and the depreciation function is convex.

Mechanism: A future increase in productivity makes future investment attractive. With investment adjustment costs, the household/firm smooths investment by increasing it immediately. Higher investment today lowers the shadow value of installed capital relative to consumption, making current utilization more attractive. More utilization raises the marginal product of labor. If the short-run wealth effect is weak, hours worked rise.

5 How the model generates comovement

5.1 Step-by-step response to good news

Consider a news shock at time 1: agents learn that productivity will rise permanently at time 3.

1. **Expected future productivity rises.** The economy anticipates a future increase in either neutral TFP A_t or investment-specific technology z_t.
2. **Investment increases today.** Adjustment costs make it costly to wait and jump investment only when the shock materializes.
3. **Marginal q falls.** The value of installed capital relative to consumption falls because capital is becoming cheaper to build.
4. **Utilization rises.** Lower replacement cost makes it attractive to run existing capital harder.
5. **Labor demand rises.** Higher utilization raises the marginal product of labor.
6. **Hours rise if wealth effects are weak.** The model avoids the standard fall in labor supply.
7. **Consumption can rise at the same time.** The economy produces more today, so it can support higher consumption and investment together.

5.2 Why each ingredient is necessary

- **Without variable utilization:** news changes expectations but has little immediate effect on output capacity. Investment can rise only by crowding out consumption.
- **Without investment adjustment costs:** agents have less reason to increase investment before the future shock occurs.
- **Without weak short-run wealth effects:** hours fall, destroying aggregate comovement.

Interpretation: The paper's logic is not that news mechanically causes booms. News causes booms only when the model contains enough internal amplification and the labor supply does not respond like a pure wealth-effect object in the short run.

6 Two-sector model and sectoral comovement

The two-sector version separates consumption-goods production and investment-goods production. The resource constraints become

$$C_t = A_t z_t^c (u_t^c K_t^c)^{1-\alpha} (N_t^c)^{\alpha},$$

$$I_t^c + I_t^i = A_t z_t^i (u_t^i K_t^i)^{1-\alpha} (N_t^i)^{\alpha}.$$

Each sector has its own capital stock, utilization rate, and investment. Total labor is allocated across sectors:

$$N_t^c + N_t^i = N_t.$$

6.1 The Christiano-Fitzgerald problem

In a two-sector neoclassical model, a shock that raises productivity in one sector can draw labor and investment away from the other sector. That generates sectoral reallocation rather than sectoral comovement. The paper shows that KPR-style preferences make it especially hard for labor in both sectors to rise together.

6.2 Why weak short-run wealth effects matter even more in two sectors

With GHH-like preferences, total labor can expand enough for both sectors to increase employment. This is why the two-sector news results require γ to be extremely close to zero. Table 1 shows that sectoral comovement under news shocks is much more demanding than aggregate comovement in the one-sector model.

Takeaway: The paper's sectoral result is the stringent test. A model that only generates aggregate comovement may still fail if sectoral labor and investment move in opposite directions.

7 Calibration and simulation design

7.1 Benchmark calibration

The benchmark one-sector model sets log utility ($\sigma = 1$), discount factor $\beta = 0.985$, labor share $\alpha = 0.64$, labor-supply parameter $\theta = 1.4$, and $\gamma = 0.001$, which makes preferences close to GHH in the short run. The investment adjustment-cost curvature is set to 1.3, and the elasticity of the marginal depreciation cost of utilization is set to 0.15.

7.2 Stochastic simulations

For business-cycle simulations, the paper focuses on investment-specific technical progress. The log of z_t follows a random walk:

$$\log z_{t+1} = \log z_t + \epsilon_{t+1}.$$

The innovation is approximated by a two-state Markov chain using data on the relative price of investment from 1947:I to 2004:IV. The model is then simulated under three information environments:

- **No signal:** agents forecast using only the Markov process.
- **Noisy signal:** agents receive a signal calibrated to the Livingston survey's forecast precision.
- **Perfect signal:** agents perfectly observe the future innovation two periods ahead.

7.3 Business-cycle moments

The simulated series are HP-filtered, and the model is compared to US moments for output, hours, investment, and consumption. The noisy-news version produces procyclical comovement and relative volatilities similar to postwar US data, even though it accounts for only part of the level of output volatility.

Economic message: The model's business-cycle properties are robust to how much news agents receive. This is important because the authors want news shocks to be feasible drivers, not a fragile assumption that destroys standard business-cycle facts.

8 Original figures and tables with interpretation

8.1 Figure 1: Investment, Earnings Growth Forecasts, and Realized Earnings; Figure 2: One-Sector Model, Response to News Shocks

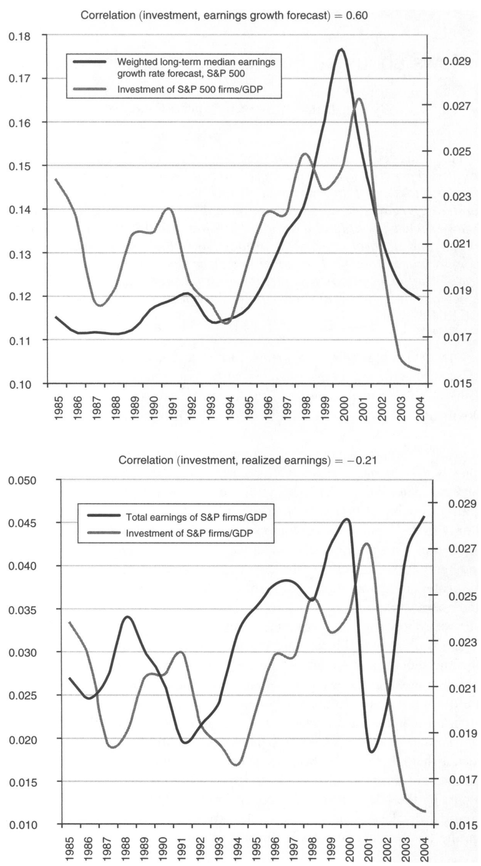


Figure 1 – Investment, Earnings Growth Forecasts, and Realized Earnings

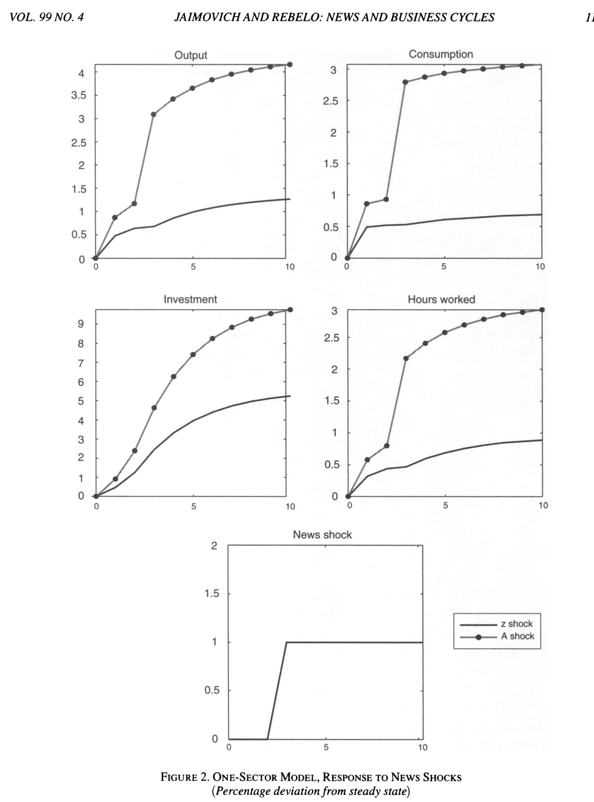


Figure 2 – One-Sector Model, Response to News Shocks

Read: **Figure 1** compares investment of S&P 500 firms to analysts' long-run earnings-growth forecasts and realized earnings. The striking fact is that investment rises with forecasts, not with realized earnings. **Figure 2** gives the one-sector impulse response to news that either A_t or z_t will rise two periods later.

Interpretation: In Figure 1, the 1990s boom looks like an expectations-driven investment episode. In Figure 2, the model turns this idea into a formal mechanism: output, consumption, investment, and hours all rise before the actual productivity improvement occurs.

Takeaway: The paper is not merely saying that expectations matter; it is showing that expectations can move all key aggregates in the same direction in a disciplined equilibrium model.

8.2 Table 1: Robustness Analysis; Figure 3: Wealth Effects on the Labor Supply of a 1 Percent Permanent Real Wage Increase

TABLE 1—ROBUSTNESS ANALYSIS

One-sector model	News A	News z
Maximum γ	0.650	0.400
Minimum adjustment costs, $\varphi''(1)$	0.370	0.400
Minimum elasticity of labor supply $(1/(\theta-1))$	0.111	0.111
Maximum elasticity of utilization	2.500	5.000

Two-sector model	Contemporaneous shocks			News shocks		
	A	z^c	z^i	A	z^c	z^i
Maximum γ	0.600	1.000	0.110	0.009	0.006	0.006
Minimum adjustment costs, $\varphi''(1)$	0.010	0.010	0.010	1.100	1.000	1.100
Minimum elasticity of labor supply $(1/(\theta-1))$	0.256	0.001	1.000	1.667	1.667	1.667
Maximum elasticity of utilization	infinity	infinity	2.800	0.300	0.300	0.250

Table 1 – Robustness Analysis

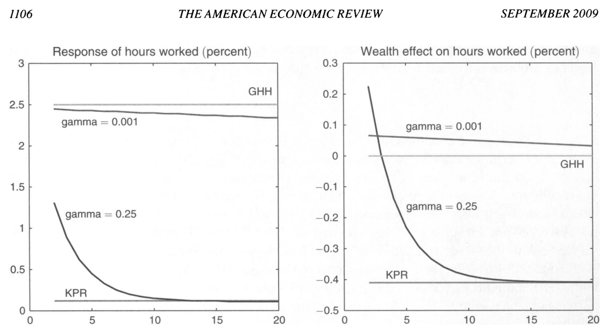


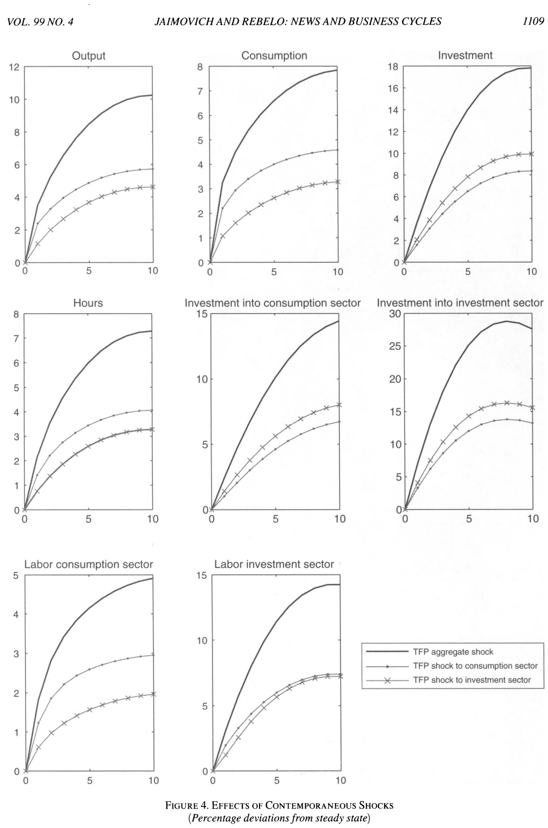
Figure 3 – Wealth Effects on the Labor Supply of a 1 Percent Permanent Real Wage Increase

Read: **Table 1** reports parameter ranges that preserve comovement. In the one-sector model, news about A_t or z_t permits moderate values of γ. In the two-sector model, especially under news shocks, admissible γ values are extremely close to zero. **Figure 3** shows how the labor response to a permanent wage increase changes as preferences move from GHH toward KPR.

Interpretation: The table and figure jointly identify the critical parameter. Aggregate comovement can survive some short-run wealth effect, but robust sectoral comovement requires almost GHH-like behavior. The labor supply must not contract when future wealth rises.

Takeaway: If you remember one calibration lesson, remember this: sectoral news-driven booms require very weak short-run labor-supply wealth effects.

8.3 Figure 4: Effects of Contemporaneous Shocks; Figure 5: Effects of News Shocks



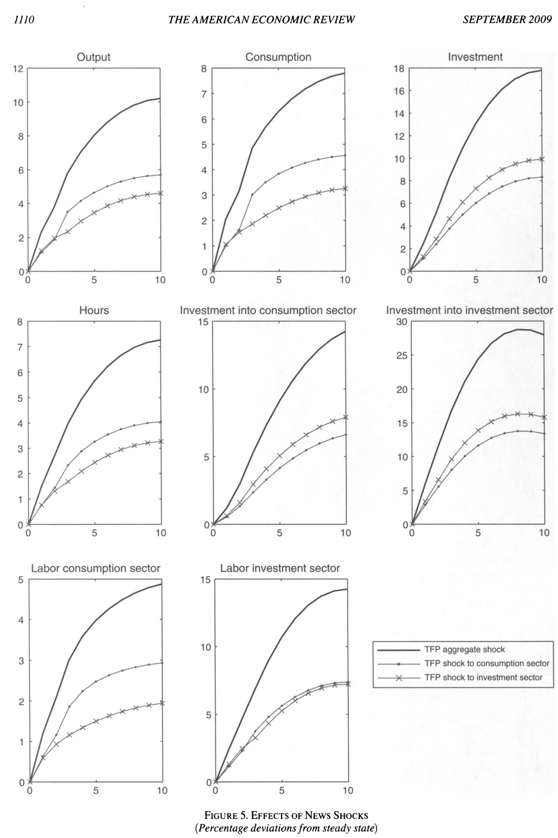


Figure 4 – Effects of Contemporaneous Shocks

Figure 5 – Effects of News Shocks

Read: Figure 4 shows responses to contemporaneous 1 percent shocks: aggregate TFP, TFP in the consumption sector, and TFP in the investment sector. **Figure 5** repeats the exercise for news shocks that arrive today but materialize two periods later.

Interpretation: Both figures show aggregate and sectoral comovement. Output, consumption, investment, total hours, sectoral investment, and sectoral labor generally rise together. This is the paper's main success relative to the standard two-sector model.

Mechanism: The responses are strongest when investment-specific or investment-sector news changes the expected value of accumulating capital. Adjustment costs pull investment forward, utilization amplifies production, and weak wealth effects permit labor to rise in both sectors.

8.4 Table 2: Business Cycle Moments; Figure 6: Average Recession in the Model and in US Data

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TABLE 2—BUSINESS CYCLE MOMENTS

	Data			Model		
	1947–2004	1947–1983	1983–2004	No signal	Signal with Livingston-survey precision	Perfect signal
Standard deviation output	1.56	1.88	0.97	1.10	1.00	0.94
Standard deviation hours	1.51	1.88	1.00	0.78	0.71	0.67
Standard deviation investment	4.84	5.41	3.69	3.45	3.33	3.30
Standard deviation consumption	1.11	1.22	0.75	0.81	0.75	0.73
Correlation output and hours	0.86	0.88	0.87	1.00	1.00	1.00
Correlation output and investment	0.89	0.75	0.91	0.97	0.93	0.85
Correlation output and consumption	0.77	0.68	0.75	0.94	0.92	0.89

Table 2 – Business Cycle Moments

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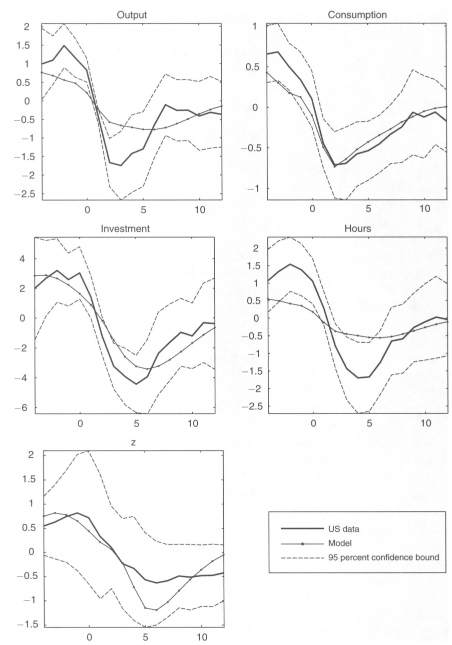

FIGURE 6. AVERAGE RECESSION IN THE MODEL AND IN US DATA

Figure 6 – Average Recession in the Model and in US Data

Read: **Table 2** compares postwar US business-cycle moments to the model under no signal, a Livingston-survey-precision signal, and a perfect signal. **Figure 6** compares average US recessions to average model recessions.

Interpretation: The model reproduces key qualitative moments: output, hours, consumption, and investment are procyclical, investment is more volatile than output, and consumption is less volatile than output. Figure 6 shows that recessions can arise even though investment-specific technological progress is not negative.

Takeaway: Recessions in this model are not caused by bad current technology. They arise when the economy receives disappointing news about future productivity growth.

9 Main results to remember

1. **News shocks can generate booms.** Good news about future fundamentals raises current output, consumption, investment, and hours in the authors' model.
2. **The standard model fails because of labor-supply wealth effects.** If households strongly value leisure after receiving good news about future wealth, hours fall and comovement fails.
3. **Three ingredients work jointly.** Variable utilization, investment adjustment costs, and weak short-run wealth effects are complements, not substitutes.

4. **Sectoral comovement is stricter than aggregate comovement.** It requires very weak short-run wealth effects and enough amplification for labor and investment to rise in both sectors.

5. **Investment-specific technology is central in simulations.** The stochastic business-cycle exercise uses investment-specific technical progress and noisy signals calibrated to forecast surveys.

6. **More information can lower volatility.** A better ability to forecast future shocks can reduce output volatility and raise persistence, offering a complementary explanation for the Great Moderation.

7. **Recessions can be disappointment episodes.** The model's recessions occur when current growth is not necessarily bad, but expected future growth worsens.

10 How to explain the paper in one paragraph

Jaimovich and Rebelo ask whether news about future fundamentals can generate realistic business-cycle fluctuations. In a standard model, good news raises wealth and lowers labor supply, so output often falls. They show that adding variable capital utilization, investment adjustment costs, and preferences with weak short-run wealth effects changes the answer. News raises investment today because adjustment costs make smoothing desirable; higher investment lowers the shadow value of installed capital and raises utilization; higher utilization increases labor demand; and weak wealth effects allow hours to rise. The result is aggregate and sectoral comovement in response to both contemporaneous and anticipated productivity shocks. In simulations with investment-specific technical progress and noisy Livingston-style signals, the model produces recognizable business-cycle moments and recessions driven by bad news about the future rather than bad current technology.

11 Useful exam-style questions

1. Why does a standard neoclassical model often generate a recession in response to good news about future productivity?

2. What role does variable capital utilization play in turning news into an immediate output response?

3. Why do investment adjustment costs make investment rise before the future shock arrives?

4. What is the economic interpretation of the parameter γ in the Jaimovich-Rebelo preference specification?

5. Why is sectoral comovement harder to generate than aggregate comovement?

6. How does Table 1 reveal the importance of weak short-run wealth effects?

7. In the model simulations, why can recessions occur even when investment-specific technological progress is never negative?

12 Limitations and interpretation cautions

- **News is difficult to measure directly.** Figure 1 and the Livingston-survey calibration are suggestive, but they do not identify structural news shocks in the data.
- **The result depends heavily on preferences.** The model's success requires short-run wealth effects on labor supply to be weak, especially in the two-sector setting.
- **The impulse responses are model-generated.** The paper is a theory and simulation contribution, not a reduced-form empirical test of news shocks.
- **Firm value issue.** With the benchmark adjustment-cost specification, good news can reduce marginal q and may reduce firm value unless decreasing returns or other mechanisms are introduced.
- **Other shocks are absent.** The model abstracts from oil shocks, monetary shocks, financial frictions, and many other disturbances that likely matter for actual recessions.

13 Bottom line

- **Core message.** News can drive business cycles if the economy responds immediately through investment and utilization, and if labor supply does not fall because of wealth effects.
- **What is new.** The paper gives a unified model that passes the aggregate and sectoral comovement tests for both contemporaneous and anticipated shocks.
- **Why it matters.** The paper makes expectations a plausible primitive driver of macro fluctuations without abandoning real business-cycle discipline.